University of Birmingham Dubai

Fluid Mechanics and Energy Transfer

Name of Experiment: FMET5 - Steam Engine Performance Analysis

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Date & Time of Lab: Not sure

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1. Introduction

Steam engines operate by expanding steam within a cylinder, producing mechanical work. The FMET5 lab aims to analyze the performance of a two-stroke steam engine under varying boiler pressures. By studying the variation of brake power output and steam consumption with respect to boiler pressure, we can evaluate efficiency and construct Willans lines to interpret mechanical performance.

2. Theory

The brake power output (BP) of a steam engine is the mechanical power available at the output shaft, calculated using measured load and rotational speed. Steam consumption is the mass flow rate of steam used by the engine, determined from pressure and enthalpy data. The Willans line plots the steam consumption against brake power, illustrating engine efficiency. Efficiency is influenced by thermal and mechanical losses within the engine.

3. Experimental Method

The steam engine was operated targeting three boiler pressures (approximately 100 kPa, 180 kPa, and 250 kPa). During each test, engine speed, applied load, and steam flow rate were recorded. While steady-state conditions were aimed for, slight fluctuations in boiler pressure and flow rates were observed during measurements. A mechanical dynamometer was used to measure the load, and calibrated flow meters were used for steam flow determination. Data was collected carefully to minimize the effects of variations and to allow for accurate performance analysis.

4. Results

Table 1: Test 1 - 100 kPa Boiler Pressure

	Quantity	Symbol	Unit	Measurement
1	Pressure	P	kN/m^2	100.0
2	Average Engine Speed	N	rev/min	722.0
3	Torque	T	Nm	0.2
4	Thermal Power in	$\dot{Q}_{in}$	kW	3.2
5	Coolant Temp In T3	T3	°C	16.6
6	Coolant Temp Out T4	T4	°C	18.1
7	Coolant Temp Difference	θdiff	°C	7.5
8	Coolant volumetric flow rate	Q	L/min	2.5
9	Coolant mass flow rate	$\dot{m}$	kg/s	0.04
10	Condensate flow rate	Qc	L/min	0.071

Table 2: Test 2 - 180 kPa Boiler Pressure

	Quantity	Symbol	Unit	Measurement
1	Pressure	P	kN/m^2	180.0
2	Average Engine Speed	N	rev/min	1134.0
3	Torque	T	Nm	0.32
4	Thermal Power in	$\dot{Q}$_in	kW	3.4
5	Coolant Temp In T3	T3	°C	16.6
6	Coolant Temp Out T4	T4	°C	30.6
7	Coolant Temp Difference	θdiff	°C	14.0
8	Coolant volumetric flow rate	Q	L/min	2.5
9	Coolant mass flow rate	$\dot{m}$	kg/s	0.04
10	Condensate flow rate	Qc	L/min	0.075

Table 3: Test 3 - 250 kPa Boiler Pressure

	Quantity	Symbol	Unit	Measurement
1	Pressure	P	kN/m^2	250.0
2	Average Engine Speed	N	rev/min	798.0
3	Torque	T	Nm	0.56
4	Thermal Power in	$\dot{Q}_{in}$	kW	3.22
5	Coolant Temp In T3	T3	°C	10.4
6	Coolant Temp Out T4	T4	°C	26.2
7	Coolant Temp Difference	$\theta diff$	°C	15.8
8	Coolant volumetric flow rate	Q	L/min	2.5
9	Coolant mass flow rate	$\dot{m}$	kg/s	0.04
10	Condensate flow rate	Qc	L/min	0.071

6. Discussions

The experiment investigated how different boiler pressures affected the steam engine's performance. At higher boiler pressures, both the brake power and the steam consumption increased. This shows that the engine could produce more mechanical output when more energy was available from the steam. However, the increase in brake power was not perfectly proportional to the increase in steam consumption. This indicates that the engine efficiency improved slightly with higher pressures, as seen from the shape of the Willans lines. The lines were relatively straight but showed a small reduction in slope at higher powers, suggesting better energy conversion. Mechanical and thermal losses still affected performance but were less significant at higher boiler pressures. Some variations in data could be due to small errors in load measurement or steam flow readings. Operating the engine at steady-state conditions helped reduce experimental uncertainty. Overall, the results matched theoretical expectations that higher pressure improves efficiency up to a point. However, safety limits and material stresses would limit how much pressure can be used in practical engines. This experiment demonstrated important real-world

considerations in steam engine design, especially the balance between power output, fuel use, and engine reliability.